

# SHOVAN SHAKYA

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🐙 [momoisgoodforhealth.github.io/](https://github.com/momoisgoodforhealth)

## Skills

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**Languages:** C, C++, Python, Verilog

**Software Development Tools:** Zephyr RTOS, OpenCV, CUDA, Vivado, ROS2, CMake, Git

**CAD Tools:** Altium, KiCAD

**Hardware & Debug Tools:** Oscilloscope, Logic Analyzer, JTAG, SWD

## Experience

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### REEKON Tools

**Boston, Massachusetts**

*Embedded Software Engineer*

*June 2025 - Current*

- Developed Zephyr RTOS firmware in C for the T1R Range hybrid measuring system on a Nordic nRF54 MCU, building the Kickstarter prototype firmware and maturing it toward production readiness for a product with \$300K+ in pre-orders.
- Implemented peripheral drivers (UART for LDM (Laser Distance Measure), stepper motors, I2C for battery charger, GPIO expander, magnetic sensor, IMU, SPI for TFT LCD).
- Improved SPI display driver refresh rate from 8 Hz to 30 Hz using semaphore based synchronization for tear-free LCD updates; implemented adaptive brightness control via ALS feedback while maintaining readability.
- Optimized T1R Range power draw across active, idle, and sleep modes via low-power firmware states and peripheral sequencing, display voltage rail tuning, reducing power draw from 112 mW to 61 mW.
- Extended BLE firmware with Bluetooth HID keyboard output and a custom GATT notify characteristic to stream measurements and real-time IMU accelerometer telemetry.
- Implemented dual-axis leveling from accelerometer data with 30 Hz update rate and LPF for stable visualization.
- Developed an Embedded Linux + STM32 based LDM scanning prototype integrating camera sensors, stepper motors and quadrature encoder feedback for planar measurements.
- Developed multithreaded C++ image processing and motor control pipelines to detect laser dots, steer stepper motor to target using field oriented control (FOC) with PID based position control.
- Achieved sub-second inverse kinematics solver (previously 30 seconds) by engineering a custom C++ math engine to replace Python bottlenecks, leveraging Pybind11 and Eigen for zero copy memory integration, SIMD hardware acceleration.
- Developed Linux kernel device driver for LDM module.

### Jaycon Systems

**Palm Bay, Florida**

*Firmware Engineering Intern*

*September 2024 - May 2025*

- Contributed to a Zynq-7000 SoC image processing and state estimation system for a NASA SBIR project, implementing connected component labeling (CCL) in C++ on ARM Cortex-A9 (15 Hz at 720p) and developing FPGA-accelerated pipelines using C++ HLS and SystemVerilog with dual-port BRAM integration.
- Enhanced STM32 production firmware in C by mitigating SPI EMI issues for FCC compliance, adding ADC based temperature logging, and implementing SPI double buffering to eliminate DMA buffer corruption.
- Exposed sensor data (thermistor, VIN, GPIO) to a desktop application via C++ with SWIG generated C# bindings for diagnostics and monitoring.

### NBCUniversal

**Orlando, Florida**

*Sensor Fusion Intern*

*May 2024 - August 2024*

- Developed STM32 firmware in C for IMU interfacing over I2C, communicating with other STM32 modules via CANBUS, and transfer IMU data to a computer using UDP via Ethernet SPI.
- Designed a 4-layer STM32-based PCB (2cm x 2cm) with BNO088 IMU, Ethernet SPI module and TJA1051 CAN IC, supporting up to 30V input voltage.
- Developed a Python tool for synchronized pose and camera data capture on a NVIDIA Jetson based ROS2 platform.

### Monterey Bay Aquarium Research Institute

**Moss Landing, California**

*Computer Vision Intern*

*June 2023 - August 2023*

- Developed OpenCV based realtime disparity and distance estimation tool for fisheye stereo cameras with accuracy range of 5 meters (+- 300 mm) for desktop and VR.
- Used multiprocessing to perform disparity and tracking in parallel.
- Integrated FathomNet YOLOv5 deep sea organism tracking model with distance estimation.
- Used Sockets to send realtime frames, and distance information between Python OpenCV application and Unity VR.

## Education

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### University of South Florida

**Tampa, Florida**

*Bachelor of Science in Electrical Engineering*

*May 2025*